

salesforce

Create Your Low-Code, Free, Open Source CI Solution

David Reed,
Lead Member of Technical Staff, Salesforce



David Reed

Lead Member of Technical Staff, Salesforce
@aoristdual / d.reed@salesforce.com



SEA Dreamin

12th & 13th November 2021



Proudly sponsored by Platinum: **accenture**

 **FormAssembly**

**U
S
T**

 **validity**

Gold:  **communicat - o**

Supported by **CAPITAL**
MARKETING & EVENTS





SEA DREAMIN

SPONSORS

Platinum


accenture

 **FormAssembly**

U ■
S T

 **validity**

Gold

 **APPISTOKI**
Virtualise Your Techscape

 **communicat** - 

Supported by:

CAPITAL 
MARKETING & EVENTS

 **salesforce**





Forward-Looking Statements



This presentation contains forward-looking statements about the Company's financial and operating results, which may include expected GAAP and non-GAAP financial and other operating and non-operating results, including revenue, net income, earnings per share, operating cash flow growth, operating margin improvement, expected revenue growth, expected current remaining performance obligation growth, expected tax rates, stock-based compensation expenses, amortization of purchased intangibles, shares outstanding, market growth, environmental, social and governance goals, expected capital allocation, including mergers and acquisitions, capital expenditures and other investments, expectations regarding closing contemplated acquisitions and contributions from acquired companies. The achievement or success of the matters covered by such forward-looking statements involves risks, uncertainties and assumptions. If any such risks or uncertainties materialize or if any of the assumptions prove incorrect, the Company's results could differ materially from the results expressed or implied by the forward-looking statements it makes. The risks and uncertainties referred to above include those factors discussed in Salesforce's reports filed from time to time with the Securities and Exchange Commission, including, but not limited to: risks associated with our ability to successfully integrate Slack Technologies, Inc.'s operations; our ability to realize the anticipated benefits of the Slack Technologies, Inc. transaction; the impact of Slack Technologies, Inc.'s business model on our ability to forecast revenue results; disruption from the transaction making it more difficult to maintain business and operational relationships; the impact of, and actions we may take in response to, the COVID-19 pandemic, related public health measures and resulting economic downturn and market volatility; our ability to maintain service performance and security levels meeting the expectations of our customers, and the resources and costs required to avoid unanticipated downtime and prevent, detect and remediate performance degradation and security breaches; our ability to secure and costs related to data center capacity and other infrastructure provided by third parties; our reliance on third-party hardware, software and platform providers; the effect of evolving domestic and foreign government regulations, including those related to the provision of services on the Internet, those related to accessing the Internet, and those addressing data privacy; current and potential litigation involving us or our industry, including litigation involving acquired entities such as Tableau; regulatory developments and regulatory investigations involving us or affecting our industry; our ability to successfully introduce new services and product features, including any efforts to expand our services beyond the CRM market; the success of our strategy of acquiring or making investments in complementary businesses and strategic partnerships; our ability to compete in the market in which we participate; the success of our business strategy and our plan to build our business; our ability to execute our business plans; our ability to continue to grow unearned revenue and remaining performance obligation; the pace of change and innovation in enterprise cloud computing services; the seasonal nature of our sales cycles; our ability to limit customer attrition and costs related to those efforts; the success of our international expansion strategy; the demands on our personnel and infrastructure resulting from significant growth in our customer base and operations; our dependency on the development and maintenance of the infrastructure of the Internet; our real estate and office facilities strategy and related costs and uncertainties; fluctuations in, and our ability to predict, our operating results and cash flows; the variability in our results arising from the accounting for term license revenue products; the performance and fair value of our investments in complementary businesses through our strategic investment portfolio; our ability to protect our intellectual property rights; our ability to develop our brands; the valuation of our deferred tax assets and the release of related valuation allowances; uncertainties regarding our tax obligations in connection with potential jurisdictional transfers of intellectual property; uncertainties regarding the effect of general economic conditions; and risks related to our debt and lease obligations.

Continuous Integration

A whirlwind tour

What is it?

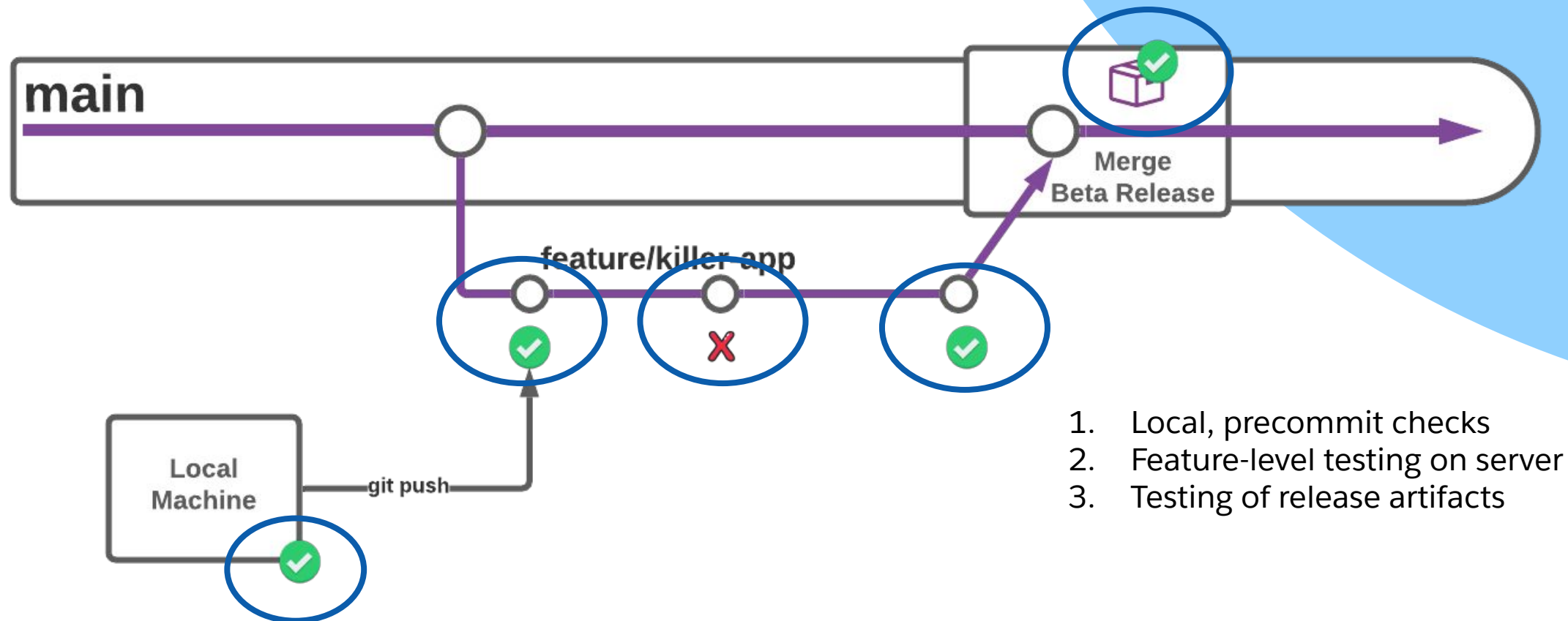
- Using automation to ensure software quality
- Run tests at every stage of the SDLC
- Scale development horizontally, safely

What's the value proposition?

- Lower risk
- Faster feedback
- Ship more value sooner

Mapping the development cycle

Note key testing points



1. Local, precommit checks
2. Feature-level testing on server
3. Testing of release artifacts

Goals



Building an open source 2GP package with a low- or no-code CI solution

On the Server

- Run Apex unit tests
- Run Jest LWC tests
- Run analysis tools
- Check formatting and linting
- Create package versions
- Install and test package versions

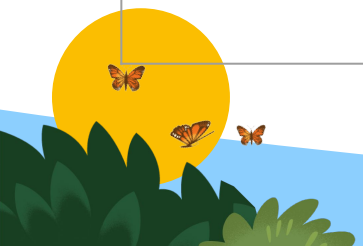
On Our Machines

- Format and lint code at commit time
- Bring valuable automation from CI into our development process

What can we test and validate?



Language	Tests	Formatting	Linting and Static Analysis	Code Coverage
Apex	✓	✓ (Prettier)	✓ (PMD)	✓
JavaScript (LWC/Aura)	✓ (Jest)	✓ (Prettier)	✓ (ESLint)	✓
XML/YAML/JSON	N/A	✓ (Prettier)	N/A	N/A
• All (End-to-End Testing)	✓ (Robot)	N/A	N/A	N/A



Let's dive in!

salesforce



Recap: Value Proposition

Shorter feedback loops, higher quality

Safe, parallel, fully automated testing

- No shared environments or bottlenecks
- Rapid iteration and feedback

Built on standard free and/or open source tools

- No cost for open source
- Free tier for closed source

Next steps

- Report code coverage to Codecov for insights
- Optimize scratch org consumption
- Pin SFDX version and cache for speed



Resources



Repository for this talk: github.com/davidmreed/sea-dreamin

Salesforce Developer Blog

- Using Salesforce DX with GitHub Actions
- How to Work with Developer Sandboxes and the Salesforce CLI
- Make the Most of Your DX Projects With Built-in Node.js Scripts

Sample Apps: <https://developer.salesforce.com/code-samples-and-sdks>

CumulusCI documentation: <https://cumulusci.readthedocs.io/en/stable/>



Thank You